

Chassis Shape Menu

What Shape Should Our Robot Base Have?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different frame shapes and imagine how each one could support the robot's job.
- **[Red: Critical Thinking & Problem Solving]** We compare trade-offs before building.
- **[Blue: Effective Communication]** We explain our design choice in the engineering portfolio.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and materials.

Decision Being Made

Your team is choosing the basic shape of the robot chassis. The chassis is the base of the robot. It holds the wheels, motors, battery, electronics, and future mechanisms such as a shooter, hopper, trigger, or intake.

A chassis is not just "the bottom of the robot." Its shape affects:

- how the robot drives,
- how stable the robot is,
- how much space is available inside,
- where balls or artifacts can move,
- where motors and electronics can go,
- how easy the robot is to repair,
- and how easy it will be to add mechanisms later.

Background Sources

Use ideas from:

- chassis design discussion,
- wheel and drive system notes,
- borrowed materials expectations,
- FIRST / FTC values,
- your team's strategy ideas,
- and any robot driving observations from Day 1.

Why This Decision Matters

Teams often want to start building immediately, but chassis shape is one of the most important early robot decisions. A strong, simple frame may drive well but leave less room for game pieces. An open frame may support an intake or pushing strategy but may need extra support to stay rigid. A narrow frame may park or share space more easily but may leave less room for electronics and mechanisms.

There is no perfect shape. Your job is to choose a shape that is good enough to build, test, and improve.

How to Use This Menu

- Read the chassis shape options.
- Discuss what each shape helps and what it makes harder.
- Think about where wheels, motors, battery, electronics, and future mechanisms might go.
- Choose a first chassis shape.
- Sketch your shape from the top view.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: O-Shape / Box Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The frame is closed on all four sides, like a rectangle or box. It may look like an "O" from the top.

What it helps: This is often strong, simple, and easier to square. It gives the robot a clear outside frame and can protect internal parts.

Best for: Teams that want a reliable driving base, simple construction, and clear places to mount motors, wheels, battery, and electronics.

Ease of use: Easy

Likelihood of success: High

Time to implement: Fast to moderate

Material needs: Usually needs enough channel or extrusion to make a full outside frame. It may require corner brackets, plates, or inside connectors.

Possible trade-offs: A closed front may block balls, artifacts, or intake paths. It may be harder to push game pieces through the robot. Future mechanisms may need to be mounted above or outside the frame.

Evidence we might watch: Does the frame stay square? Does it drive straight? Is there enough room inside for battery, electronics, motors, and mechanisms?

Option B: H-Shape Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The frame has two side rails and a cross member in the middle, like the letter "H."

What it helps: An H-shape can leave open space at the front and back. This may help with intakes, pushing balls, or interacting with game pieces from either side.

Best for: Teams that want openings at both ends of the robot or want to leave the front and back flexible for future mechanisms.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material needs: Needs two strong side rails and at least one cross member. The team may need brackets or plates to keep the frame rigid.

Possible trade-offs: The frame may twist more than a closed box frame. The middle cross member may block the path for a ball, hopper, battery, or electronics. The team must decide what the open spaces are for.

Evidence we might watch: Does the frame twist during turns? Can balls or artifacts move through the open spaces? Does the cross member block anything important?

Option C: A-Shape / One-End-Open Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: The frame is closed on one end and open on the other. It is like an H-shape where one end is closed. In this course, "A-shape" means one end is open for interaction with game pieces; it does not need to be angled.

What it helps: The open end can guide balls or artifacts, make space for an intake, or allow the robot to push game pieces toward the human operator or scoring area. The closed end can add strength and give the team a stable place for battery, electronics, or a shooter mount.

Best for: Teams that want one clear "front" of the robot for collecting, pushing, or interacting with game pieces.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: Needs side rails, a rear cross member, and possibly extra brackets or plates to reduce twisting at the open end.

Possible trade-offs: The open end may be less rigid. The team must clearly define the robot's front and protect wheels, wires, and mechanisms near the opening.

Evidence we might watch: Does the open end help with balls or artifacts? Does the frame twist? Can the team still mount wheels, motors, electronics, and battery safely?

Option D: Narrow Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot is intentionally built narrower than the maximum allowed size.

What it helps: A narrow robot may fit through tighter spaces, share the field with other robots, park more easily, or move around crowded areas.

Best for: Teams that value maneuverability, parking, passing, or working near other robots.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material needs: May use shorter channels or extrusions, but internal space becomes limited.

Possible trade-offs: Less room for battery, electronics, shooter, hopper, trigger, or intake. The robot may be less stable if it becomes tall or top-heavy.

Evidence we might watch: Can the robot still fit all required parts? Does it tip during turns? Does the narrow shape improve parking or movement?

Option E: Wide / Full-Size Frame

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot uses most of the allowed footprint.

What it helps: A wider frame gives more room for mechanisms, electronics, battery, and stable mounting. It may be easier to build and more stable.

Best for: Teams that need room for mechanisms and want a stable base.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Moderate

Material needs: May require longer channels or extrusions. Long pieces may be limited.

Possible trade-offs: A wide robot may have trouble sharing space, parking with other robots, or moving through tight areas.

Evidence we might watch: Does the extra space help the build? Does the size create problems with parking, turning, or crowding?

Shape Comparison Table

Shape	Best Feature	Main Risk	Good For
O-shape / box	Strong and simple	May block game-piece paths	Reliable driving base
H-shape	Open front and back	May twist or block middle path	Flexible intake/pushing options
A-shape / one-end-open	One clear open end	Open end may need support	Front intake, pushing, or guiding
Narrow frame	Easier to fit in tight spaces	Less room inside	Parking, passing, shared field space
Wide / full-size frame	Stable and roomy	Harder to share space	Mechanisms and stability

Materials Note

Your team may use parts from different systems, including **goBILDA**, **REV**, and **Tetrix**. These systems may use different hole patterns, sizes, connectors, and fasteners.

You may have access to:

- U-channel from goBILDA, REV, or Tetrix,
- REV 15 mm extrusion,
- brackets,
- plates,

- inside connectors,
- bolts, nuts, and spacers.

You will not have unlimited access to every length or every part. Your chassis shape must work with the materials that are actually available.

This is part of engineering. Real teams design under constraints.

Chassis Shape Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Chassis shape				
Robot width				
Open spaces				
Cross members / support				

Sketch Requirement

Draw a top-view sketch of your chassis plan.

Your sketch should label:

- front of robot,
- back of robot,
- frame shape,
- drive wheels,
- omni wheels,
- motor locations,
- battery location,
- electronics location,
- open space for balls, artifacts, intake, shooter, hopper, or trigger.

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis shape and draw a labeled top-view sketch.
Should Do	Explain why the shape fits your team's strategy and name at least one trade-off.

Level	Expectation
Could Do	Compare two chassis shapes, explain how each would affect scoring or future mechanisms, and choose using evidence or measurements.

SLD Connection

Main SLD Prompt

Should our chassis be simple and strong, or open and flexible for future mechanisms?

Supporting Questions

- What does our robot need to do first?
- Where might balls or artifacts need to move?
- Where will the battery and electronics go?
- What shape gives us the best chance of building a working robot quickly?
- What shape leaves room for future mechanisms?
- What material constraints might change our first choice?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis shape,
- labeled top-view sketch,
- explanation of why the team chose the shape,
- trade-off statement,
- materials the team expects to use,
- evidence the team will watch during testing.

Reflection

Our chosen chassis shape is:

We chose this shape because:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Chassis Material Menu

What Should We Build the Frame From?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different building systems and ways to solve the same design problem.
- **[Red: Critical Thinking & Problem Solving]** We compare material trade-offs before building.
- **[Green: Personal Development]** We make responsible choices based on time, skill, tools, and available materials.
- **[Orange: Cultural Understanding & Civic Engagement]** We use shared and borrowed materials responsibly.
- **[Blue: Effective Communication]** We explain our material choice in the engineering portfolio.

Decision Being Made

Your team is choosing what materials or building system to use for the robot chassis. The chassis shape matters, but the material also matters. Different systems have different strengths, fasteners, hole patterns, sizes, connectors, and tools.

Your team may not have unlimited access to every part or every length. This is normal. Real engineering teams design with the materials, money, time, and tools they actually have.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Physics and Space Planning**
- **Borrowed Equipment and Community Trust**
- materials available in the classroom,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

The frame material affects:

- how strong the chassis is,
- how easy it is to build,
- how easy it is to add attachments,
- how much space the frame uses,
- how easy the robot is to repair,
- how well parts from different systems fit together,

- how much time the team spends on the base instead of scoring mechanisms,
- and how responsibly the team uses shared or borrowed materials.

There is no perfect material choice. Your team should choose the material that best fits your design, skill level, available parts, and time.

How to Use This Menu

- Review the material options.
- Compare strength, ease of use, time, and attachment options.
- Check what materials are actually available.
- Decide whether your team should build a new chassis, modify an existing chassis, or recycle a known chassis.
- Record the choice, trade-off, and evidence plan in the engineering portfolio.

Menu Options

Option A: goBILDA U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: goBILDA U-channel is a rigid metal channel with many holes for attaching brackets, motors, wheels, shafts, and other parts.

What it helps: It is strong, precise, and designed for robotics. It works well for sturdy frames and reliable mounting.

Best for: Teams that want a strong chassis with many attachment points.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: goBILDA channel, compatible screws, nuts, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Specific lengths may not be available. Some parts may not line up cleanly with REV or Tetrix parts. Working inside channel can become hard after the robot is assembled.

Evidence we might watch: Does the frame stay rigid? Are attachments easy to add? Can we reach screws, motors, battery, and electronics?

Option B: REV U-Channel

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: REV U-channel is a metal channel used in the REV building system. It provides structure and attachment points.

What it helps: It can create strong frame members and may work well with REV brackets, plates, shafts, motors, and electronics.

Best for: Teams using mostly REV parts or needing a simple channel-based frame.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Material / tooling needs: REV channel, compatible fasteners, brackets, plates, hubs, shafts, and hex tools.

Possible trade-offs: Hole patterns and dimensions may not match goBILDA or Tetrix parts perfectly. Teams may need adapters or creative brackets.

Evidence we might watch: Can we attach wheels, motors, and mechanisms securely? Do parts line up well enough without forcing them?

Option C: REV 15 mm Extrusion

Graduate Expectation Connection: [Purple: Curiosity & Creativity]

What it means: REV 15 mm extrusion is a smaller beam-like structure that allows parts to be placed in many positions along the rail.

What it helps: Extrusion can be flexible for mounting because parts can often slide or be placed continuously instead of only at fixed holes. It is compact and useful for adjustable structures.

Best for: Teams that need flexible attachment points, small structures, supports, or adjustable mounts.

Ease of use: Medium to hard

Likelihood of success: Medium

Time to implement: Moderate to slow

Material / tooling needs: REV extrusion, brackets, corner connectors, compatible screws, nuts, and tools.

Possible trade-offs: It may be less obvious for beginners to assemble. It may require special connectors. A full chassis made only from extrusion may need careful bracing to stay square and rigid.

Evidence we might watch: Can we attach parts where we want? Does the frame stay square? Does anything slide, loosen, or twist during driving?

Option D: Tetrix Channel or Mixed Channel Parts

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Tetrix and other channel parts may be available. Teams may combine parts from different systems when the exact preferred material is not available.

What it helps: Mixed systems can make use of available materials and may solve a parts shortage.

Best for: Teams that are flexible, willing to test fit, and able to use brackets or plates to connect parts from different systems.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Tetrix channel, compatible fasteners, adapters, brackets, plates, and possibly parts from multiple systems.

Possible trade-offs: Different systems may not line up cleanly. Holes, screws, shafts, hubs, and spacing may not match. Teams may spend extra time solving connection problems.

Evidence we might watch: Are connections secure? Are parts being forced together? Does the frame stay aligned after driving?

Option E: Hybrid Frame

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team intentionally combines systems. For example, the main chassis might use goBILDA channel, while smaller adjustable mounts use REV extrusion.

What it helps: A hybrid frame can use the strengths of different systems. Strong channel can support the base, while extrusion or brackets can help with adjustable mechanisms.

Best for: Teams that have a clear reason for mixing systems and enough patience to manage fit issues.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Material / tooling needs: Parts from multiple systems, adapter plates or brackets, compatible fasteners, and careful test fitting.

Possible trade-offs: Mixing systems can create alignment, spacing, and fastener problems. It may take more time than using one system.

Evidence we might watch: Does each system solve a specific problem? Are mixed connections strong and repairable?

Option F: Recycle or Modify a Known Chassis

Graduate Expectation Connection: [Green: Personal Development], [Orange: Cultural Understanding & Civic Engagement], and [Red: Critical Thinking & Problem Solving]

What it means: Instead of building a brand-new chassis from scratch, the team uses an existing chassis, known base, or partially built robot frame and modifies it for the challenge.

What it helps: This can save time, reduce risk, and give the team more time for driving, programming, shooter design, hopper/trigger work, testing, and portfolio development.

Best for: Teams that want a reliable starting point, have limited build time, or need to focus on scoring mechanisms and controls.

Ease of use: Easy to medium

Likelihood of success: High if the chassis is in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, repair tools, possible replacement screws, wheels, hubs, motors, or brackets.

Possible trade-offs: The team may have less ownership over the original design. The existing chassis may not perfectly fit the team's strategy. Old choices may limit where attachments, battery, electronics, or game-piece paths can go.

Evidence we might watch: Does the existing chassis drive reliably? What design limits does it create? How much time did it save? What did the team modify?

Comparison Table

Option	Best Feature	Main Risk	Good For
goBILDA U-channel	Strong and precise	Specific lengths/parts may be limited	Strong chassis base
REV U-channel	Works well with REV ecosystem	May not align with other systems	REV-based builds
REV 15 mm extrusion	Flexible placement	Needs careful connectors/bracing	Adjustable mounts and compact structures
Tetrix / mixed channel	Uses available parts	Fit and fastener problems	Teams solving material shortages
Hybrid frame	Uses strengths of multiple systems	More complex connections	Teams with a clear mixed-system plan
Recycled known chassis	Saves time and reduces risk	Less original design control	Teams prioritizing mechanisms, code, and testing

Materials Availability Check

Before finalizing your material choice, check:

- Do we have enough pieces in the lengths we need?
- Do we have the correct fasteners?
- Do we have brackets or connectors that fit?
- Do we have the correct hubs, shafts, spacers, and wheels?
- Can we attach motors securely?
- Can we reach screws later for repair?
- Are we using borrowed materials responsibly?
- Do we need to request a limited part?

Material Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Main frame material				
Connector/bracket plan				
Mixed systems?				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Recycled chassis?				

Must / Should / Could

Level	Expectation
Must Do	Choose a chassis material or existing chassis and explain why it is good enough to start building.
Should Do	Explain the main trade-off of the material choice and check whether the needed parts are available.
Could Do	Compare two material systems or compare new-build vs. recycled chassis using time, reliability, repair access, and future mechanism space.

SLD Connection

Main SLD Prompt

Should our team build a new chassis from preferred materials, adapt to the materials available, or recycle a known chassis to save time?

Supporting Questions

- What material gives us the best chance of a reliable driving base?
- What material gives us the most flexibility for future attachments?
- What material is actually available?
- How much time should we spend on the chassis before moving to mechanisms?
- When is recycling a known chassis a smart engineering decision?
- What do we lose if we do not build the chassis from scratch?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected chassis material or existing chassis,
- reason for the choice,
- material trade-off statement,
- parts availability notes,
- sketch or photo of planned frame,
- and evidence the team will watch during testing.

Reflection

Our chassis material choice is:

We chose this because:

The main trade-off we accepted is:

One material constraint we need to manage is:

Motor Placement Menu

Where Should the Drive Motors Go?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different ways to fit motors into the chassis.
- **[Red: Critical Thinking & Problem Solving]** We compare turning, space, attachment access, and repair trade-offs.
- **[Green: Personal Development]** We make responsible choices based on available parts, time, and team skill.
- **[Blue: Effective Communication]** We explain motor placement using sketches, evidence, and trade-offs.

Decision Being Made

Your team is deciding where to place the drive motors. Motor placement affects how the robot turns, how easy it is to control, how much open space remains in the chassis, how easy it is to attach future mechanisms, and how easy the robot is to repair.

The drive motors are not just power sources. Their location shapes the whole robot layout.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Wheel Selection Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available motors, hubs, gears, shafts, and brackets,
- and your team's chassis sketch.

Why This Decision Matters

Motor placement is one of the hidden decisions that affects everything else.

Motors in the center may make the robot easier to turn, but they take up valuable middle space. Motors near the ends may leave the center open, but the robot may need more room to turn. Motors inside channels may look clean and protected, but they can make it harder to add attachments later.

Your team needs to decide what matters most: turning, space, protection, repair access, future mechanisms, or speed of building.

How to Use This Menu

- Review the motor placement options.
- Check your chassis shape and wheel layout.
- Think about where the battery, electronics, shooter, hopper, and trigger might go.

- Choose a first motor placement plan.
- Sketch the motors on your top-view chassis drawing.
- Record your choice, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: Motors Near the Center

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The drive motors and powered wheels are placed closer to the middle of the robot.

What it helps: This can make the robot easier to turn and may help it rotate in place more smoothly. It can make the robot feel more controllable for a human driver.

Best for: Teams that want easier turning, predictable control, and a robot that can line up without needing a huge turning space.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, compatible wheels, hubs, shafts, brackets, spacers, and space in the middle of the chassis.

Possible trade-offs: Motors in the center occupy valuable middle space. This may interfere with a ball path, hopper, shooter mount, battery, electronics, or future intake.

Evidence we might watch: Does the robot turn smoothly? Can it rotate in place? Is there still enough room for the battery, electronics, and future mechanisms?

Option B: Motors Near the Ends

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The motors and powered wheels are placed closer to the front and/or back ends of the robot.

What it helps: This can leave the center of the robot more open for game pieces, electronics, battery, hopper, or mechanisms.

Best for: Teams that need the middle of the robot open for a ball path, hopper, shooter, or intake.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material / tooling needs: Motor mounts, wheels, hubs, shafts, brackets, and wiring that can reach the motors safely.

Possible trade-offs: The robot may be harder to turn. It may need more space to rotate, and the driver may need more practice to control it.

Evidence we might watch: Does the robot need too much room to turn? Does it line up accurately? Does the open middle space help with future mechanisms?

Option C: Motors Along the Side Rails

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Motors are mounted along the left and right sides of the chassis, usually near the wheels but not necessarily inside the channel.

What it helps: This can make motors easier to see, wire, repair, and adjust. It may keep the center more open while still supporting a simple drive layout.

Best for: Teams that want repair access and a clear, understandable drivetrain.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Fast to moderate

Material / tooling needs: Side mounting brackets, motor mounts, wheels, hubs, spacers, and wire management.

Possible trade-offs: Motors may stick out or be more exposed to contact. They may reduce space for side attachments.

Evidence we might watch: Can we reach the motors and wires? Are the motors protected enough? Do they block future attachments?

Option D: Motors Inside the Channels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The motors are mounted inside U-channel or tucked tightly into the frame. This may use compact motor placement, bevel gears, or special brackets.

What it helps: This can protect motors and create a clean, compact drivetrain. It may keep the outside of the robot clear.

Best for: Teams that want a compact design and have enough time, tools, and support to build carefully.

Ease of use: Hard

Likelihood of success: Medium to risky

Time to implement: Slow

Material / tooling needs: Compatible channel, motor mounts, gears or shafts if needed, careful alignment, hex tools, and possibly mentor support.

Possible trade-offs: Once motors are inside the channel, it can be much harder to attach new parts to that channel. Repairs may also be harder. If the motor or gear alignment is wrong, debugging can take a long time.

Evidence we might watch: Can we still attach parts to the channel? Can we reach screws and wires? Does the drivetrain run smoothly without binding?

Option E: Motors Protected but Accessible

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The team places motors where they are partly protected by the frame but still easy enough to reach for repair, wiring, and inspection.

What it helps: This balances durability with repair access. It avoids hiding the motors so deeply that they become difficult to fix.

Best for: Teams that want a practical, reliable course robot.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, brackets, protective frame members, enough space for tools and wires.

Possible trade-offs: The robot may not look as compact as a hidden-motor design, and motors may still take up useful space.

Evidence we might watch: Can the team fix a loose motor or wire quickly? Are motors protected during normal driving and contact?

Option F: Keep Motor Placement From a Recycled Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: If the team is using a known or recycled chassis, it keeps the existing motor placement unless there is a clear reason to change it.

What it helps: This saves time and reduces the risk of rebuilding the drivetrain incorrectly.

Best for: Teams that want to spend more time on programming, driving, shooter design, hopper, trigger, testing, and portfolio work.

Ease of use: Easy

Likelihood of success: High if the drivetrain already works

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, replacement screws or wires if needed.

Possible trade-offs: The existing motor placement may limit chassis shape, open space, turning behavior, or future mechanisms.

Evidence we might watch: Does the recycled drivetrain drive reliably? What space does it leave or block? Would changing it cost more time than it saves?

Comparison Table

Option	Best Feature	Main Risk	Good For
Motors near center	Easier turning and rotation	Uses middle space	Driver control and tight turns
Motors near ends	Opens middle space	Harder turning	Game-piece paths and mechanisms
Motors along side rails	Easy access and clear layout	More exposed	Repairable simple drivetrains
Motors inside channels	Compact and protected	Hard to attach/repair later	Advanced compact builds
Protected but accessible	Balanced practical design	Still uses space	Reliable course robot

Option	Best Feature	Main Risk	Good For
Keep recycled placement	Saves time	Limits design choices	Fast reliable start

Motor Placement Questions

Before finalizing motor placement, answer:

- Where are the powered wheels?
- Can the robot turn in the space available?
- Can the driver control the robot easily?
- Does motor placement block the center of the robot?
- Is there room for the battery and electronics?
- Is there room for the shooter, hopper, trigger, or intake?
- Can we reach motor screws and wires later?
- Are motors protected from normal contact?
- Are wires safe from wheels and moving parts?
- Does this placement fit our chassis shape and materials?

Motor Placement Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Motor placement				
Powered wheel location				
Space affected				
Repair access				

Must / Should / Could

Level	Expectation
Must Do	Choose where the drive motors will go and show them on a labeled chassis sketch.
Should Do	Explain how motor placement affects turning, driver control, space, and repair access.
Could Do	Compare two motor placements using turning radius, driver feedback, or available space measurements.

SLD Connection

Main SLD Prompt

Should our drive motors be placed for easier turning, more open space, better protection, or easier repair?

Supporting Questions

- Is turning control more important than open space?
- What future mechanisms might need the middle of the chassis?
- How much repair access do we need?
- Should motors be protected even if they become harder to reach?
- What motor placement best fits our chassis shape and wheel choice?
- When is a clean compact design worth the extra build difficulty?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected motor placement,
- labeled chassis sketch showing motors and powered wheels,
- explanation of why the team chose this placement,
- trade-off statement,
- repair/access notes,
- and evidence the team will watch during testing.

Reflection

Our motor placement choice is:

We chose this because:

The main trade-off we accepted is:

The evidence we will watch during testing is:

Wheel Selection Menu

How Should Our Robot Touch the Field?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different wheel types and imagine how they affect robot movement.
- **[Red: Critical Thinking & Problem Solving]** We compare traction, turning, speed, control, and reliability.
- **[Green: Personal Development]** We make responsible choices based on available materials and team skill.
- **[Blue: Effective Communication]** We explain wheel choices using sketches, test data, and observations.

Decision Being Made

Your team is choosing the wheels for the chassis. Wheels affect how the robot drives, turns, accelerates, pushes, stops, and handles contact with the field or game objects.

Teams may have access to wheels from different systems, including Tetrix, REV, goBILDA, and others. Wheel sizes, tread patterns, rubber density, hubs, axles, and connectors may all be different.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available wheel bins,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

Wheels are not just "the things that roll." They determine how the robot interacts with the floor.

Wheel choice affects:

- traction,
- turning,
- speed,
- stability,
- pushing ability,
- driver control,
- autonomous consistency,

- clearance,
- and how easily the wheels attach to motors, hubs, shafts, and the chassis.

A wheel that looks good may not fit the shaft. A wheel that grips well may make turning harder. A wheel that turns easily may not push well. Your team needs to choose based on the job the robot needs to do.

How to Use This Menu

- Review the wheel options.
- Check which wheels are actually available.
- Check whether the wheels fit your hubs, shafts, axles, and chassis.
- Decide which wheels are powered and which wheels support the robot.
- Sketch the wheel layout.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Wheel Design Factors

Wheel Size

Larger wheels can move the robot farther with each rotation, which can make the robot faster. They can also increase clearance off the floor. Smaller wheels may be easier to fit and may give more control.

Larger Wheels	Smaller Wheels
More distance per rotation	More compact
More floor clearance	Often easier to fit
May increase speed	May feel easier to control
May need more torque	May reduce clearance

Tread and Rubber

Soft rubber and grippy tread can help the robot push and accelerate. Harder wheels or smoother wheels may slip more but can sometimes turn more easily.

More Grip	Less Grip
Better pushing and acceleration	Easier turning in some layouts
More control when driving straight	Less strain during turns
May resist sideways movement	May slip under load
Can make turning harder	May reduce pushing power

Fit and Connection

Before choosing a wheel, check:

- Does the wheel fit the hub?

- Does the hub fit the shaft or axle?
- Is the shaft round or hex?
- Is the set screw tight?
- Is the wheel supported?
- Does the wheel rub the frame?
- Do we have spacers to stop side-to-side sliding?
- Does the wheel match the motor placement and chassis shape?

Menu Options

Option A: Standard Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Traction wheels are regular wheels designed to grip the floor.

What it helps: They provide pushing power, acceleration, and strong forward/backward movement.

Best for: Drive wheels powered by motors.

Ease of use: Easy to medium

Likelihood of success: High if they fit the hub and shaft

Time to implement: Fast to moderate

Material / tooling needs: Wheels, hubs, shafts/axles, set screws, spacers, bearings or supports, and hex tools.

Possible trade-offs: Too much grip can make turning harder, especially if all four wheels are traction wheels.

Evidence we might watch: Does the robot drive straight? Does it turn smoothly? Do the wheels slip or scrub during turns?

Option B: Omni Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Omni wheels have small rollers around the edge. They roll forward like normal wheels, but can slide sideways more easily.

What it helps: Omni wheels reduce sideways resistance during turns. A common beginner layout uses two powered traction wheels and two omni support wheels.

Best for: Helping a two-motor robot turn more smoothly.

Ease of use: Medium

Likelihood of success: High if installed correctly

Time to implement: Moderate

Material / tooling needs: Omni wheels, compatible hubs, shafts/axles, spacers, and supports.

Possible trade-offs: Omni wheels usually provide less sideways resistance, so the robot may be easier to push sideways. They may provide less traction than regular wheels.

Evidence we might watch: Does the robot turn more easily? Does it drift sideways? Does it still drive straight enough for testing?

Option C: Four Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: All four wheels are regular traction wheels.

What it helps: This can give the robot strong grip and pushing power.

Best for: Robots that need to push or hold position and do not need to turn very tightly.

Ease of use: Easy

Likelihood of success: Medium

Time to implement: Fast

Material / tooling needs: Four matching traction wheels, hubs, axles/shafts, spacers, and supports.

Possible trade-offs: The robot may fight itself while turning because all four wheels resist sideways sliding. This is called wheel scrub. It can make turning jerky, slow, or hard on motors.

Evidence we might watch: Does the robot turn smoothly? Do motors strain during turns? Does the robot jerk or skid?

Option D: Two Traction Wheels + Two Omni Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Two powered traction wheels move the robot. Two omni wheels support the robot and help it turn.

What it helps: This is a strong beginner layout because it balances powered movement with easier turning.

Best for: Two-motor course robots that need simple, reliable driving.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Two traction wheels, two omni wheels, compatible hubs/shafts/spacers, and supports.

Possible trade-offs: Less pushing power than four traction wheels. The robot may slide more easily from the omni-wheel side.

Evidence we might watch: Does the robot turn predictably? Does it drive straight? Can the driver control it easily?

Option E: Mecanum Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Mecanum wheels have angled rollers that allow a robot to drive forward/backward, turn, and move sideways when used in a four-motor drivetrain.

What it helps: Mecanum drive can make it easier to line up with goals, move around field objects, and strafe sideways.

Best for: Advanced teams with enough motors, wheels, time, and programming support.

Ease of use: Hard

Likelihood of success: Risky for a short course unless the base already exists

Time to implement: Slow

Material / tooling needs: Four mecanum wheels, four drive motors, correct wheel orientation, motor controllers, more complex code, tuning, and driver practice.

Possible trade-offs: More expensive, more complex, more programming, more wiring, and more debugging. It may take time away from shooter, hopper, trigger, portfolio, and testing.

Evidence we might watch: Can the robot strafe reliably? Do all wheels spin in the correct direction? Can the driver control it under pressure?

Option F: Reused Wheels From an Existing Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The team uses wheels already mounted on a known chassis or salvages wheels from an existing robot base.

What it helps: This can save time and reduce fit problems because the wheels, hubs, shafts, and supports may already work together.

Best for: Teams that want to prioritize robot reliability, programming, mechanisms, or testing time.

Ease of use: Easy to medium

Likelihood of success: High if the wheels are in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis or wheel assemblies, inspection checklist, hex tools, replacement screws, spacers, or hubs if needed.

Possible trade-offs: The wheel layout may limit chassis shape, motor placement, ground clearance, or future mechanisms.

Evidence we might watch: Do the wheels spin freely? Are set screws tight? Does the robot drive straight and turn reliably?

Wheel Layout Questions

Before finalizing the wheel choice, answer:

- Which wheels are powered by motors?
- Which wheels are support wheels?
- Where are the omni wheels, if used?
- Do the wheels fit the hubs and shafts?
- Are the set screws tight?
- Is the wheel supported on the axle?
- Does the wheel rub the frame?

- Is there enough clearance off the floor?
- Will the robot turn smoothly?
- Will the driver be able to control the robot?

Wheel Assembly Checklist

Check	Yes / No
Wheel fits the hub	
Hub fits the shaft or axle	
Set screw is tight	
Wheel spins freely	
Wheel does not rub the frame	
Spacers stop side-to-side sliding	
Axle or shaft is supported	
Left and right drive wheels are aligned	
Omni wheels roll in the correct direction	
Robot has enough floor clearance	

Comparison Table

Option	Best Feature	Main Risk	Good For
Standard traction wheels	Grip and pushing	Turning may be harder	Powered drive wheels
Omni wheels	Easier turning	Less sideways resistance	Support wheels on simple drivetrains
Four traction wheels	Strong grip	Wheel scrub during turns	Pushing or holding position
Two traction + two omni	Balanced beginner drive	Less pushing than four traction	Reliable two-motor robot
Mecanum wheels	Sideways movement	Complex, expensive, needs four motors	Advanced competition-style driving
Reused wheel assemblies	Saves time	Limits design choices	Reliable fast start

Wheel Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Drive wheels				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Support wheels				
Wheel size				
Wheel material/tread				
Wheel layout				

Must / Should / Could

Level	Expectation
Must Do	Choose wheels that fit the chassis and identify which wheels are powered.
Should Do	Explain how the wheel choice affects turning, traction, control, and reliability.
Could Do	Compare two wheel layouts using a drive test, turning test, or driver feedback.

SLD Connection

Main SLD Prompt

Should our robot prioritize traction, easy turning, speed, or driver control when choosing wheels?

Supporting Questions

- What wheel choice gives us the best chance of driving reliably?
- When is more grip helpful?
- When can too much grip make turning harder?
- Should we use omni wheels to make turning easier?
- Is mecanum worth the extra complexity in a short course?
- Should we reuse a known wheel setup to save time?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected wheel types,
- powered wheel locations,
- support wheel locations,
- wheel layout sketch,
- wheel assembly checklist,

- explanation of the main trade-off,
- and evidence the team will watch during testing.

Reflection

Our wheel choice is:

We chose this because:

The main trade-off we accepted is:

The evidence we will watch during testing is:
